Physics 5A Discussion (Week 4)

Department of Physics and Astronomy, UCLA

Consider a rocket moving in space. We set up the coordinate system such that the rocket starts its motion at the origin, with the x-axis representing horizontal motion and the y-axis representing vertical motion. Here is some space for you to doodle :)

(a) From time $t = 0\,\text{s}$ to $t = 0.37\,\text{s}$, the rocket moves to the left and stops at a distance of $0.285\,\text{m}$ to the left of the origin. Write down the displacement vector of the rocket from $t = 0\,\text{s}$ to $t = 0.37\,\text{s}$. Write it in vector component form, that is $\Delta\vec{x} = (x_\text{f} - x_\text{i}\,,\, y_\text{f} - y_\text{i})$.

(b) At time $t = 0.37\,\text{s}$, the rocket starts moving to the right. At time $t = 0.54\,\text{s}$, the rocket ends at a distance $0.2\,\text{m}$ to the left of the origin. Write down the displacement vector of the rocket from $t = 0.37\,\text{s}$ to $t = 0.54\,\text{s}$, and write down the position vector of the rocket at $t = 0.54\,\text{s}$. In terms of their physical meaning, what is the difference between these two vectors?

(c) Using the definition of the displacement vector $\Delta\vec{x} = (x_\text{f} - x_\text{i}\,,\, y_\text{f} - y_\text{i})$, write down the displacement vector for the rocket's journey from $t = 0\,\text{s}$ to $t = 0.54\,\text{s}$. This is different from the displacement vector from part (b).

(d) Now we consider a two-step process:

1. From $t = 0.54\,\text{s}$ to $t = 0.80\,\text{s}$, the rocket goes from $0.2\,\text{m}$ to the left of the origin back to the origin.

2. From $t = 0.80\,\text{s}$ to $t = 16.40\,\text{s}$, the rocket goes in the "uphill" direction on a ramp (I called it the x'-direction in the previous worksheet), making an angle 30° above the horizontal, and it travels $200\,\text{m}$ in this direction.

Find the component form of the displacement vector in each of the two processes. That is, find the displacement from $t = 0.54\,\text{s}$ to $t = 0.80\,\text{s}$ and the displacement from $t = 0.80\,\text{s}$ to $t = 16.40\,\text{s}$.

(e) Find, by adding the displacement vectors found in part (d), the total displacement from $t = 0.54\,\text{s}$ to $t = 16.40\,\text{s}$.

(f) Find the average velocity from $t = 0.54\,\text{s}$ to $t = 16.40\,\text{s}$, in vector component form.

(g) Find the total displacement from $t = 0\,\text{s}$ to $t = 16.40\,\text{s}$: first write it in vector component form; then compute its magnitude and direction (measure the direction as an angle measured counterclockwise from the positive x-axis).

(h) Find the total distance traveled by the rocket from $t = 0\,\text{s}$ to $t = 16.40\,\text{s}$.

(i) Find the average speed of the rocket from $t = 0\,\text{s}$ to $t = 16.40\,\text{s}$.

Solution to (a). The rocket ends at a point directly to the left of the origin, so we expect to see only an x-component in its displacement.

$$\Delta\vec{x} = (x_\mathrm{f} - x_\mathrm{i}\,,\; y_\mathrm{f} - y_\mathrm{i}) = (-0.285 - 0,\; 0 - 0)\,\mathrm{m} = (-0.285,\; 0)\,\mathrm{m}\,.$$

Solution to (b). The starting position at $t = 0.37\,\mathrm{s}$ is $(-0.285,\; 0)\,\mathrm{m}$, and the ending position at $t = 0.54\,\mathrm{s}$ is $(-0.2,\; 0)\,\mathrm{m}$, so

$$\Delta\vec{x} = ((-0.2,\; 0) - (-0.285,\; 0))\;\mathrm{m} = (0.085,\; 0)\,\mathrm{m}\,.$$

The x-component is positive because the rocket moves to the right from $t = 0.37\,\mathrm{s}$ to $t = 0.54\,\mathrm{s}$.

The position vector at $t = 0.54\,\mathrm{s}$ points from the origin to the final position of the rocket. In other words, it is the coordinate corresponding to the location of the rocket at $t = 0.54\,\mathrm{s}$,

$$\vec{x} = (-0.2,\; 0)\,\mathrm{m}\,.$$

The displacement vector describes the change in the rocket's position from $t = 0.37\,\mathrm{s}$ to $t = 0.54\,\mathrm{s}$, while the position vector describes where the rocket is located relative to the origin at $t = 0.54\,\mathrm{s}$.

Solution to (c). The starting position at $t = 0\,\mathrm{s}$ is the origin $(0,\; 0)\,\mathrm{m}$, and the ending position at $t = 0.54\,\mathrm{s}$ is $(-0.2,\; 0)\,\mathrm{m}$, so

$$\Delta\vec{x} = ((-0.2,\; 0) - (0,\; 0))\;\mathrm{m} = (-0.2,\; 0)\,\mathrm{m}\,.$$

The x-component is negative because the rocket ends at a point to the left of where it started.

Solution to (d). First, from $t = 0.54\,\mathrm{s}$ to $t = 0.80\,\mathrm{s}$, the rocket starts at $(-0.2,\; 0)\,\mathrm{m}$ and ends at $(0,\; 0)\,\mathrm{m}$, so

$$\Delta\vec{x}_1 = ((0,\; 0) - (-0.2,\; 0))\;\mathrm{m} = (0.2,\; 0)\,\mathrm{m}\,.$$

Then, from $t = 0.80\,\mathrm{s}$ to $t = 16.40\,\mathrm{s}$, the rocket starts at $(0,\; 0)\,\mathrm{m}$ and ends at a point $\vec{p}$ that we need to calculate. The rocket travels $200\,\mathrm{m}$, which is the hypotenuse, at an angle of 30° above the horizontal. Therefore, we can calculate the x- and y-components of $\vec{p}$ using trigonometry:

$$p_x = 200\cos(30^\circ) = 173.2\,\mathrm{m}\,, \qquad p_y = 200\sin(30^\circ) = 100\,\mathrm{m}\,.$$

Therefore, the displacement for the second step is

$$\Delta\vec{x}_2 = ((173.2,\; 100) - (0, 0))\;\mathrm{m} = (173.2,\; 100)\,\mathrm{m}\,.$$

Solution to (e). The total displacement from $t = 0.54\,\mathrm{s}$ to $t = 16.40\,\mathrm{s}$ can be calculated in two ways. The first way is by adding the two displacement vectors found in part (d),

$$\Delta\vec{x} = \Delta\vec{x}_1 + \Delta\vec{x}_2 = (0.2 + 173.2,\; 0 + 100)\,\mathrm{m} = (173.4,\; 100)\,\mathrm{m}\,.$$

The second way is to use "final minus initial":

$$\Delta\vec{x} = ((173.2,\ 100) - (-0.2, 0))\ \text{m} = (173.4,\ 100)\,\text{m}\,.$$

This is the same result as above.

Solution to (f). The displacement from $t = 0.54\,\text{s}$ to $t = 16.40\,\text{s}$ is calculated in part (e). Average velocity is a vector, calculated by dividing that displacement by the time it takes to travel, so

$$\bar{\vec{v}} = (\bar{v}_x,\ \bar{v}_y) = \left(\frac{173.4}{16.40 - 0.54},\ \frac{100}{16.40 - 0.54}\right)\ \text{m/s} = (10.9,\ 6.3)\ \text{m/s}\,.$$

Solution to (g). The displacement from $t = 0\,\text{s}$ to $t = 16.40\,\text{s}$ can, again, be calculated in two ways. The first way is to add all the displacements; the second way is to use "final minus initial". Here we take the second way:

$$\Delta\vec{x} = ((173.2,\ 100) - (0,\ 0))\ \text{m} = (173.2,\ 100)\,\text{m}\,.$$

Using the first way gives

$$\Delta\vec{x} = ((-0.285,\ 0) + (0.085,\ 0) + (0.2,\ 0) + (173.2,\ 100))\ \text{m} = (173.2,\ 100)\,\text{m}\,.$$

This is the same result as above. The magnitude of this vector is

$$|\Delta\vec{x}| = \sqrt{173.2^2 + 100^2}\,\text{m} = 200\,\text{m}\,,$$

and the direction, measured counterclockwise from the positive x-axis, is

$$\theta = \arctan\left(\frac{\Delta\vec{x}_y}{\Delta\vec{x}_x}\right) = \arctan\left(\frac{100}{173.2}\right) = 30^\circ\,.$$

Solution to (h). The total distance traveled is calculated by adding the distance in each segment regardless of its direction:

1. From $t = 0\,\text{s}$ to $t = 0.37\,\text{s}$, it moves to the left for $0.285\,\text{m}$.
2. From $t = 0.37\,\text{s}$ to $t = 0.54\,\text{s}$, it moves to the right for $(0.285 - 0.2)\,\text{m} = 0.085\,\text{m}$.
3. From $t = 0.54\,\text{s}$ to $t = 0.80\,\text{s}$, it moves to the right for $(0.2 - 0)\,\text{m} = 0.2\,\text{m}$.
4. From $t = 0.80\,\text{s}$ to $t = 16.40\,\text{s}$, it moves uphill for $200\,\text{m}$.

Thus in total,

$$d_{\text{total}} = (0.285 + 0.085 + 0.2 + 200)\,\text{m} = 200.57\,\text{m}\,.$$

Solution to (i). The average speed is the total distance over the time it takes to travel. Using the total distance traveled calculated in part (h), we obtain

$$\bar{s} = \frac{200.57}{16.40 - 0}\,\text{m/s} = 12.23\,\text{m/s}\,.$$



Shangxiajiu Pedestrian Street

A historic commercial walkway in Guangzhou, China. Pictured in December 2025.

Used to be a one of the most popular places in the city in early 2000s. Now is more deserted ...